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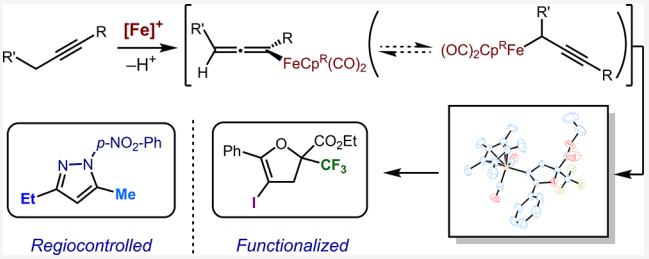
Regiocontrolled Coupling of Alkynes and Dipolar Reagents: Iron-Mediated [3 + 2] Cycloadditions Revisited

Jin Zhu,[†] Austin C. Durham,[†] Yidong Wang, James C. Corcoran, Xiao-Dong Zuo, Steven J. Geib, and Yi-Ming Wang*

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ABSTRACT: Cyclopentadienyliron dicarbonyl based allenyliron complexes were prepared, and their formal [3 + 2] cycloaddition with a number of dipolar reagents was investigated as a means of preparing heterocyclic compounds in a regiocontrolled manner. In addition, the mechanism of the isomerization of an allenyliron complex to its propargyliron tautomer was investigated. Results in support of both radical and two-electron mechanisms for isomerization are presented.



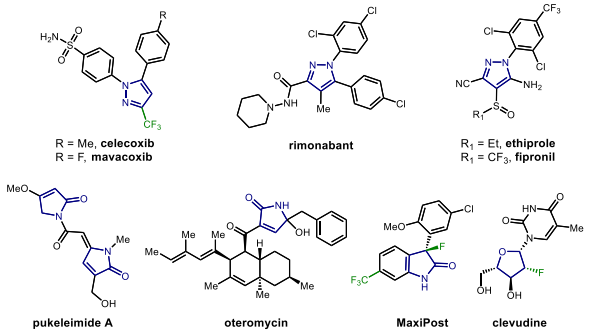
INTRODUCTION

In recent decades, the synthesis and functionalization of heterocycles have continued to be core areas of organic synthesis research due to their central role in the development of bioactive and therapeutic molecules.[1] Among these species, nitrogen-containing five-membered heterocycles have raised particular interest. *N*-Arylpyrazoles, for example, constitute key substructures in a range of molecules of pharmaceutical and agricultural interest,[2] including cyclooxygenase-2 (COX-2) inhibitors such as celecoxib, mavacoxib, and rimonabant, as well as insecticides/acaricides such as fipronil and ethiprole (Scheme 1, top).[3] Routes to access dihydropyrrole derivatives are particularly valuable within the context of recent efforts to incorporate saturated[4] and partially saturated N-heterocycles[5] into the toolkit of medicinal chemistry. Additionally, they are also valuable as intermediates in synthesis and are observed in numerous natural products and synthetic biologically active compounds (Scheme 1, bottom). Similarly, a modern surge in the use of fluorinated and trifluoromethylated targets in medicinal chemistry necessitates the continued development of methods to produce synthetically challenging fluorine-containing five-membered heterocycles.[6]

Although classical strategies for the preparation of these compounds often suffice, these approaches sometimes result in the formation of regioisomeric mixtures. Thus, the development of regioselective routes to substituted five-membered heterocycles has received particular attention.[7] Generally, the regiochemistry can be controlled by utilizing prefunctionalized substrates,[7a,b] by multicomponent processes,[7c] or through various transition-metal-catalyzed reactions.[7d] Among the approaches utilizing transition metals, methods using Co,[8a] Cu,[8b] Au,[8c] Ru,[8d] and Rh[8e] have been extensively investigated and successfully put into practice.

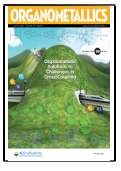
The formal cycloaddition of dipolar reagents ($E^{\delta+}$=$Nu^{\delta-}$) to unsaturated transition-metal complexes is a versatile route for the synthesis of heterocycles.[9] In many cases, transition-metal complexes, especially metal carbonyl complexes,[10] are easily handled and are useful in organic synthesis, given their straightforward preparation and good stability to air and moisture. In particular, due to their regiospecific reactivity as nucleophiles (via S_E2' addition) and latent reactivity as precursors to electrophilic cationic π-complexes, propargyl and allenyl derivatives of cyclopentadienyliron dicarbonyl complexes are particularly versatile, functioning as all-carbon

Scheme 1. N-Containing or Fluorinated Five-Membered Heterocycles in Therapeutics and Natural Products



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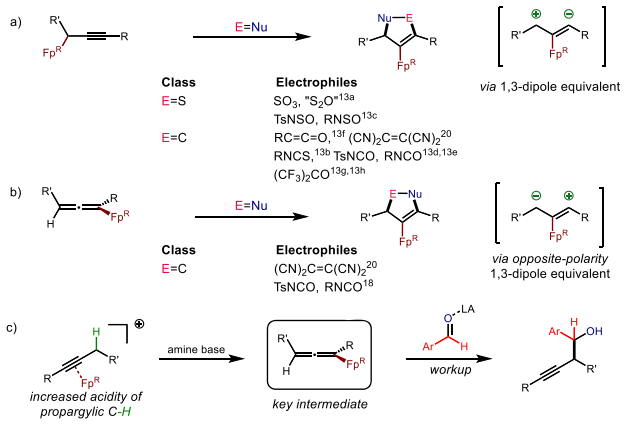
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1,3-dipoles that can be used for the synthesis of five-membered carbo- and heterocycles (Scheme 2a,b; $Fp^R = Cp^RFe(CO)_2$, Cp^R = substituted cyclopentadienyl ligand). The recently reported cycloaddition reactions of alkynes and iron-coordinated 1,3-dipolar organic fragments represents an interesting variant of this strategy.[8f,g]

Scheme 2. Synthetic Applications of Propargyl- and Allenyliron Species



Dating back to the 1970s, the first [3 + 2] cycloaddition reactions of propargyliron complexes were reported by Wojcicki and co-workers, who extended their earlier work on SO_2 insertion[11] into allylmanganese pentacarbonyl to propargyliron complexes to afford crystallographically characterized [3 + 2] cycloadducts.[12] Subsequently, the cycloadditions of transition-metal 2-alkynyl complexes with S-centered dipolar reagents, including sulfur oxides (SO_2, SO_3, masked S_2O)[13a] and *N*-sulfinylamines and *N*-sulfinylamides,[13c] as well as C-centered dipolar reagents, including isothiocyanates,[13b] isocyanates,[13d,e] electron-deficient alkenes, alkynes, and ketenes,[13f−h] were disclosed (Scheme 2a). Owing to the activation by the coordinated metal, cationic η^2-allene complexes, arising from S_E2' electrophilic attack of the acetylenic position, were proposed to undergo annulation by attack of a pendant nucleophilic moiety to deliver the observed [3 + 2] cycloadduct.[14] Detailed investigations into the mechanism have been carried out by analyzing the stereochemical outcome of the reaction with tetrasubstituted alkenes and through kinetic studies with a series of substituted isocyanate reagents.[13e]

Metal allenyl complexes are metallotropic tautomers of propargylmetal complexes.[15] The interconversion of propargyl- and allenylmetal species via a formal 1,3-sigmatropic rearrangement has been well-documented for main-group elements and a number of transition metals.[16] However, this transformation has not yet been directly observed for organoiron species, although observations made by Rosenblum and co-workers suggested this possibility.[17] In comparison with the convenient preparation of propargyliron complexes from versatile propargyl halides, the application of allenyliron species has been much less documented, presumably due to the limited number of methods for preparing these reagents. Reported allenyliron species could only be accessed by S_N2' displacement of terminal propargylic (pseudo)halides[18,19] or those with alkoxy substituents at the β-position[19] with $[CpFe(CO)_2]^-$ as the nucleophile. The allenyliron complexes were subjected to electrophiles such as tetracyanoethylene[20] and sulfonyl isocyanates,[18] giving [3 + 2] cycloaddition products isomeric with those derived from the tautomeric propargyliron species (Scheme 2b).

Recently, our group developed an iron-catalyzed α-functionalization reaction of alkynes and alkenes with activated electrophiles.[21] By drastically increasing the acidity of the propargylic/allylic C−H bond (Scheme 2c),[22] formation of a π-complex with the cationic iron fragment allows for the deprotonation of the alkene or alkyne at the α-position using a weak amine base. The resultant σ-allyliron or σ-allenyliron complex subsequently undergoes reaction with the electrophile to give the functionalized product. The key allenyliron intermediate of the catalytic process was successfully isolated in stoichiometric experiments[23,24] and crystallographically characterized.[24]

On the basis of the numerous precedents for [3 + 2] cycloaddition processes and this general approach for preparing allenyliron complexes, we hypothesized that simple alkynes and dipolar reagents could be coupled in a regioselective manner by way of allenyliron species prepared via deprotonation of cationic iron alkyne complexes. Moreover, we expected this strategy to allow for the regioselective formation of products complementary to those derived from the tautomeric propargyliron complexes.[25c] Demetalation of the product cycloadduct would furthermore provide access to a number of highly substituted heterocycles from simple starting materials. Indeed, a variety of conditions have been reported for the elaboration of vinyliron species,[9c] including protodemetalation,[13g] halodemetalation, carbonylative demetalation using single-electron oxidants,[25a,b] and nonoxidative functionalization processes using Gilman or alkyllithium reagents.[25c,d]

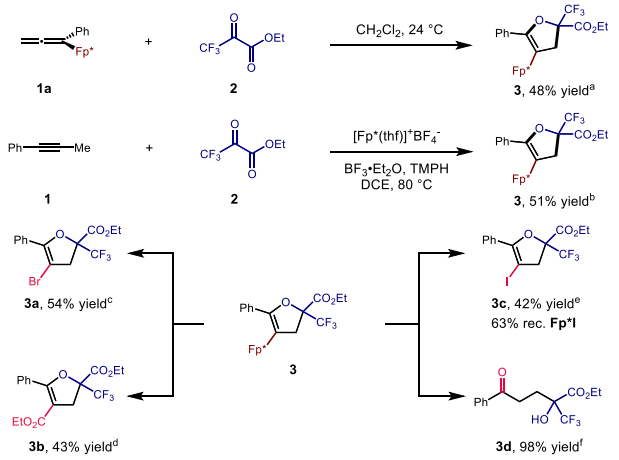
Given the limited data on the fundamental reactivity of allenyliron complexes and the synthetic interest in regioselective syntheses of five-membered heterocycles using readily accessed alkynes as starting materials, we explored the stoichiometric reactivity of these organometallic complexes with a number of electrophilic reagents. In this report, we disclose the synthesis of a collection of five-membered heterocycles resulting from these studies. In addition, during the course of these investigations, we observed the isomerization of an allenyliron complex to the tautomeric propargyliron species. We report studies in relation to the rate and mechanism of this process and consider its implications for the regioselectivity of the formal [3 + 2] cycloaddition.

RESULTS AND DISCUSSION

Carbonyl Derivatives as Cycloaddends. We first examined the reactivity of pentamethylcyclopentadienyliron dicarbonyl derived allenyliron **1a** with ethyl trifluoropyruvate (**2**) as a coupling partner, given the current interest in the preparation of trifluoromethylated heterocycles[6d] and the potential for downstream elaboration of the installed ethyl ester function. As anticipated, allenyliron **1a** and **2** underwent [3 + 2] cycloaddition under mild conditions to afford the vinyliron complex **3** as a single regioisomer (≥20:1 rr by NMR analysis of the crude material) in 48% yield. Complex **3** could also be prepared in slightly higher yield directly from 1-phenyl-1-propyne (**1**) by combining it with **2** and $[Fp^*(thf)]^+BF_4^-$ ($Fp^* = Cp^*Fe(CO)_2$) in the presence of 2,2,6,6-tetramethylpiperidine (TMPH) as base, via the *in situ* generation of **1a** (Scheme 3, top and middle). Complex **3** could be recrystallized from CH_2Cl_2/hexanes for characterization by single-

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Scheme 3. Vinyliron Dihydrofuran Synthesis and Divergent Transformations



[a]Conditions: **1a** (1.0 equiv), **2** (1.5 equiv), CH_2Cl_2, 24 °C. [b]Conditions: **1** (2.0 equiv), **2** (1.0 equiv), $[Fp^*(thf)]^+BF_4^-$ (1.0 equiv), $BF_3{\cdot}Et_2O$ (2.0 equiv), TMPH (3.0 equiv), DCE, 80 °C. [c]Conditions: **3** (1.0 equiv), *N*-bromosuccinimide (1.0 equiv), CH_2Cl_2, 24 °C. [d]Conditions: **3** (1.0 equiv), $(NH_4)_2Ce(NO_3)_6$ (1.9 equiv), K_2CO_3 (2.9 equiv), $EtOH/CH_2Cl_2$, 24 °C. [e]Conditions: **3** (1.0 equiv), I_2(1.0 equiv), CH_2Cl_2, 0−24 °C. [f]Conditions: **3** (1.0 equiv), TFA (2.0 equiv), wet CH_2Cl_2, 24 °C.

crystal X-ray diffraction (Figure 1). As a demonstration of synthetic versatility, **3** was subjected to a series of demetalation conditions to furnish functionalized organic products (Scheme 3, bottom).

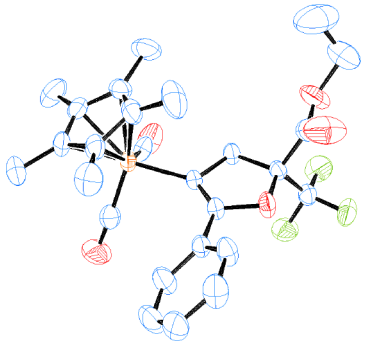
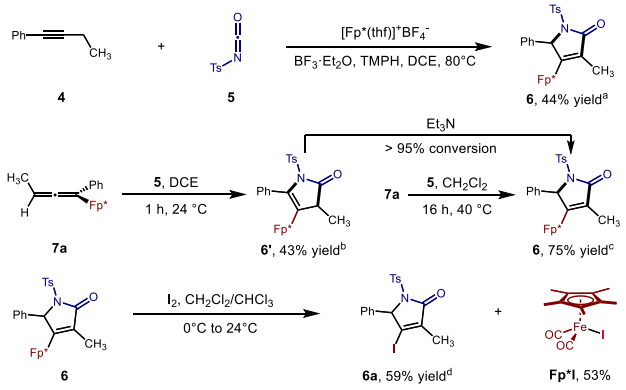


Figure 1. ORTEP (thermal ellipsoids at 50% probability) plot of the molecular structure of **3**. Hydrogen atoms are omitted for clarity.

We also investigated the synthesis of a densely substituted 1,5-dihydro-2*H*-pyrrol-2-one derivative by coupling 1-phenyl-1-butyne (**4**) with *p*-toluenesulfonyl isocyanate (**5**). Thus, in the presence of TMPH and $BF_3{\cdot}Et_2O$, **4** and **5** reacted to give the 1,5-dihydropyrrolone-derived vinyliron complex **6** (Scheme 4, top). We also investigated the preparation of **6** from the isolated allenyliron intermediate **7a**. As was observed by Rosenblum and co-workers,[17] the initially formed [3 + 2] cycloadduct after short reaction times (1 h) was vinyliron species **6′**, derived from 1,3-dihydro-2*H*-pyrrol-2-one, which could be isomerized to **6** by treatment with Et_3N or by allowing **5** and **7a** to react at higher temperature and longer reaction times (Scheme 4, middle). In the presence of I_2, **6** was transformed into **6a** in moderate yield, with Fp*I being recovered (Scheme 4, bottom). Vinyliron complex **6** could be

Scheme 4. Synthesis of Tautotomeric Dihydropyrrolone Vinyliron Complexes



[a]Conditions: **4** (2.0 equiv), **5** (1.1 equiv), $[Fp^*(thf)]^+BF_4^-$ (1.0 equiv), $BF_3{\cdot}Et_2O$ (2.0 equiv), TMPH (3.0 equiv), DCE, 80 °C. [b]Conditions: **7a** (1.0 equiv), **5** (1.1 equiv), DCE, 24 °C, 1 h. [c]Conditions: **7a** (1.0 equiv), **5** (1.1 equiv), CH_2Cl_2, 40 °C, 16 h. [d]Conditions: **6** (1.3 equiv), I_2 (1.0 equiv), $CH_2Cl_2/CHCl_3$, 0−24 °C.

crystallized from CH_2Cl_2/hexanes for characterization by single-crystal X-ray diffraction (Figure 2). In comparison to

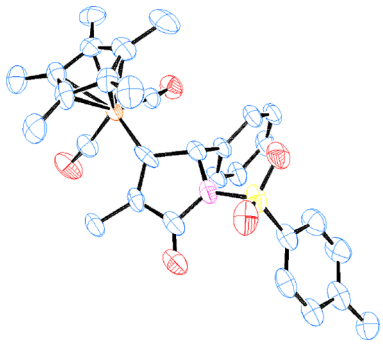


Figure 2. ORTEP (thermal ellipsoids at 50% probability) plot of the molecular structure of **6**. Hydrogen atoms are omitted for clarity.

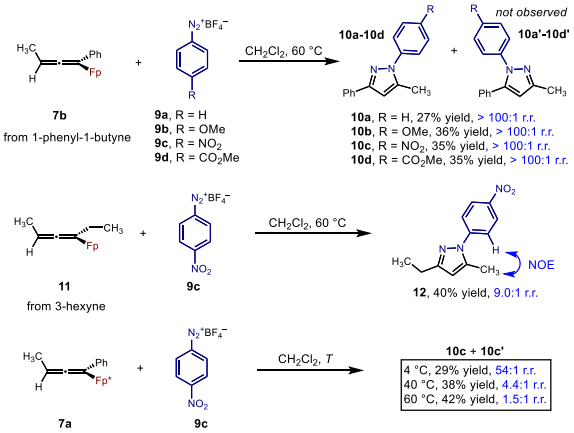
the C(sp^2)−Fe bond in **3** (2.002 Å), which is situated at the electron-rich *β*-position of an enol ether, the C(sp^2)−Fe bond in iron complex **6**, located at an electron-poor position, is shorter (1.977 Å).

Diazonium and Azo Compounds as Cycloaddends. We next turned to diazonium and azo compounds to investigate the possibility of C−N bond formation using alkyne-derived allenyliron complexes. We were particularly interested in the use of arenediazonium salts, by virtue of their simple preparation and versatile reactivity as electrophilic nitrogen sources.[26] On the basis of our previous results, we expected the [3 + 2] cycloaddition of arenediazonium tetrafluoroborates with allenyliron complexes to furnish a convenient and regioselective route to *N*-arylpyrazoles. Previously, diazonium salts have been applied to the synthesis of a number of heterocyclic scaffolds.[27] In particular, pyrazoles have previously been prepared by formal cycloaddition of diazonium salts with 1,3-enones,[28a] 1,4-enones,[28b] and cyclopropanols.[28c]

When vinyliron complex **7b** was reacted with the series of 4-substituted arenediazonium tetrafluoroborates **9a−d**, organic products identified as pyrazoles could be isolated in modest yields (27−36%, Scheme 5, top). However, upon comparison

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Scheme 5. Pyrazole Synthesis Using Cyclopentadienyl-Allenyliron Cycloaddition to Aryldiazonium Salts[a]



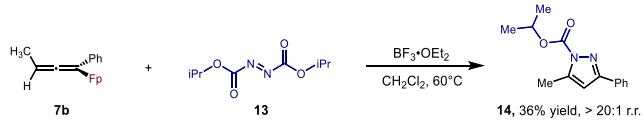
[a]Conditions: **7a**, **7b**, or **11** (1.0 equiv), **9** (1.5 equiv), CH_2Cl_2, 60 °C. Regioisomeric ratios reported are those of the crude material.

with literature characterization data of substituted pyrazoles,[29] we noticed that the regiochemical outcome was unexpected. The assigned connectivity was further confirmed by X-ray diffraction (see the Supporting Information). Regardless of the electronic nature of the substituents on the arenediazonium salt, only one regioisomer of the pyrazole was observed in the ^{1}H NMR spectrum of the crude material. Finally, we found that the dialkyl-substituted allenyliron species **11** derived from 3-hexyne also participated in the [3 + 2] cycloaddition. Notably, this route gave rise to the 3,5-dialkyl-substituted pyrazole **12** that would be challenging to prepare with good regioselectivity by conventional methods (Scheme 5, middle). Notably, while the condensation of a hydrazine and a 3-alkyn-1-one has been optimized to give a similar pyrazole in a moderate regioisomeric ratio (up to 6.8:1 rr),[30] the approach presented here provides a route to achieve a better regioselectivity from the alkyne and diazonium salt as readily available starting materials (9.0:1 rr).

Postulating that it would act as a more nucleophilic substrate, we investigated an Fp*-derived allenyl complex as the starting material. Indeed, a slight increase in yield was observed when the reaction was conducted under the same reaction conditions, but the change in ligand from Fp to Fp* also resulted in a sharp decrease in regioselectivity (from >100:1 to 1.5:1 rr for the reaction of **7a** with **9c**). The initially expected pyrazole regioisomer was observed in considerably larger quantities, albeit still as the minor isomer. It was found that the regioselectivity diminished as the temperature increased (Scheme 5, bottom). We posited that the unexpected regiochemistry could be explained by interconversion of allenyliron and propargyliron isomers, with the favored regioisomer arising from a Curtin−Hammett scenario. Though we were unable to definitively observe the propargyliron tautomer of **7a** or **7b**, suggesting an unfavorable equilibrium for tautomerization, we were able to directly study the allenyl to propargyl isomerization of the related complex **1a** (*vide infra*).

To further explore the scope of N-centered dipolar reagents that could be employed, we investigated the reaction with azodicarboxylate ester **13**, a slightly less reactive electrophile in comparison to diazonium ions.[31] Notably, Wojcicki and co-workers noted that no reaction was observed between Fp(η^1-$CH_2C{\equiv}CCH_3$) and an azodicarboxylate ester.[13c] We found, however, that when **7b** and **13** were reacted in the presence of $BF_3{\cdot}OEt_2$ as an additive, [3 + 2] cycloaddition took place with concomitant dealkoxycarbonylation to afford a moderate yield (36%) of *N*-acylpyrazole **14** as a single regioisomer (Scheme 6).[32]

Scheme 6. Pyrazole Synthesis Using Cyclopentadienyl-Allenyliron Cycloaddition to Diisopropyl Azodicarboxylate[a]



[a]Conditions: **7b** (1.0 equiv), **13** (1.5 equiv), $BF_3{\cdot}Et_2O$ (1.5 equiv), CH_2Cl_2, 60 °C.

Allenyliron to Propargyliron Isomerization. During the course of our investigations, it was found that the purity and stability of the Fp*-based allenyliron species **1a** derived from 1-phenyl-1-propyne varied greatly depending on the exact conditions of its synthesis and purification. The reaction of 1-phenyl-1-propyne with Fp*I/$AgBF_4$ followed by deprotonation with Et_3N gave a crude product mixture, consisting mainly of the expected product **1a** and a trace amount (<1%) to ca. 70% of a second iron complex, depending on the batch. A comparison of the ^{1}H NMR spectrum of this side product (δ 1.60 ppm, C_6D_6) with data reported by Welker and co-workers[19] and the $^{13}C\{^1H\}$ (δ −12.2 ppm, C_6D_6) NMR spectrum with other reported data[15a] for related propargylmetal complexes allowed its identification as propargyliron complex **1a′**. Moreover, it was found that, upon standing as a solid or a solution in CD_2Cl_2 or C_6D_6, the allenyliron complex **1a** in the crude material isomerized to propargyliron complex **1a′** to give a mixture in which the propargyliron species was predominant (propargyliron:allenyliron ≈ 92:8), suggesting equilibration between the two species.

Previously, Rosenblum and co-workers observed that cyclopentadienyliron η^1-allyl complexes bearing deuterium labels or alkyl substituents on the allyl ligand underwent 1,3-migration of the iron center to give label-scrambled or rearranged products under mild conditions (0−60 °C).[33] On the basis of a crossover experiment and the observation that label scrambling was accelerated by irradiation in the presence of substoichiometric cyclopentadienyliron dicarbonyl dimer, a source of Fp$^\bullet$, a radical chain mechanism involving homolytic allylic substitution (S_H2') was proposed. A similar process for the isomerization of propargyl/allenyliron systems by 1,3-migration of the iron center was proposed but not observed or investigated directly.[17]

Given the relevance of the isomerization to the unexpected regiochemistry of the [3 + 2] cycloaddition of iron complexes with diazonium salts described in the previous section, we set out to investigate the rate and mechanism of the observed allenyl to propargyl isomerization. First, a reliable procedure for the preparation of allenyliron complex **1a** in high chemical and isomeric purity was developed (>98% purity and >100:1 allenyliron:propargyliron by ^{1}H NMR; see the Supporting Information). When a solution of carefully purified **1a** in CD_2Cl_2 or C_6D_6 was prepared and monitored with the rigorous exclusion of light and oxygen at 24 °C, isomerization

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to **1a′** proceeded slowly, with <10% conversion taking place after 24 h (Figure 3).

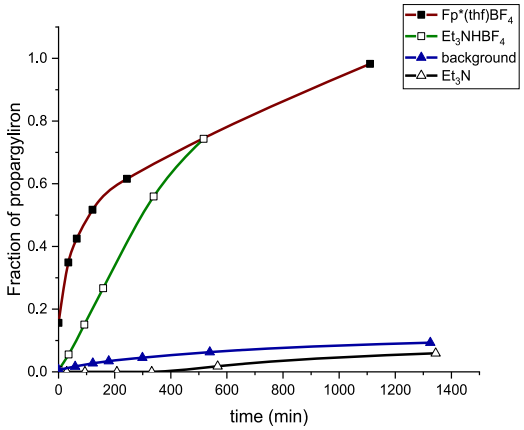
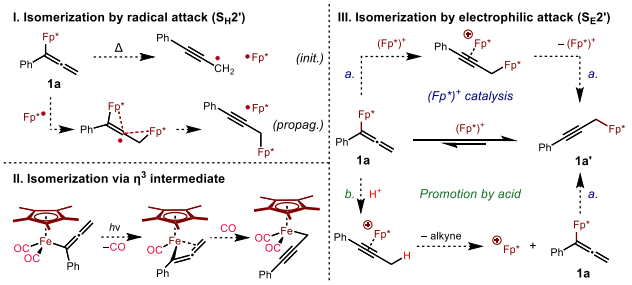


Figure 3. Fraction of η^1-propargyliron **1a′** vs time for **1a** → **1a′**.

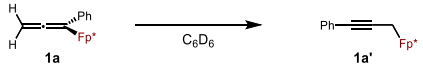
We initially hypothesized two pathways for the allenyl to propargyl isomerization: a radical (S_H2') mechanism analogous to Rosenblum's proposal for the isomerization of allyliron species and an electrophilic (S_E2') pathway, on the basis of previously reported mixed η^1,η^2 neutral/cationic bimetallic allyliron complexes, for the isomerization of **1a** to **1a′** (Scheme 7, pathways I and III).[34] To examine the likelihood of the

Scheme 7. Plausible Mechanisms for Allenyl to Propargyl Isomerization of Fp*-Iron Complexes



radical pathway I, **1a** was heated in the presence of a variety of radical traps (Table 1). In each case (entry 1 vs entries 2 and 3 and entry 4 vs entries 5 and 6), a decrease in yield was

Table 1. Radical Scavenger Effect on the η^1-Allenyliron to η^1-Propargyliron Isomerization



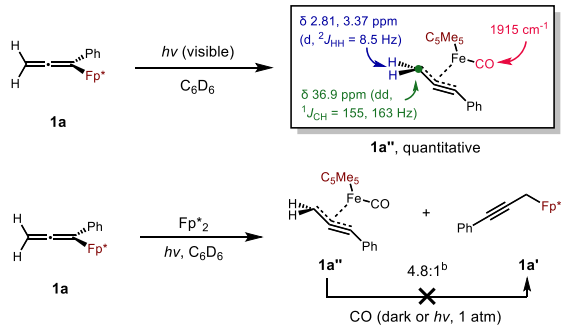
entry	radical scavenger[c]	conversion (%)
1[a]	*none*	85
2[a]	O_2 purged	64
3[a]	BzOOBz 20%	65
4[b]	*none*	60
5[b]	TEMPO 5%	7
6[b]	DHA 50%	44

[a]Conditions: **1a** (0.0125 mmol), C_6D_6 (0.5 mL), 51.0 °C, 1 h. [b]Conditions: **1a** (0.01 mmol), C_6D_6 (0.5 mL), 45.0 °C, 1 h. [c]Abbreviations: TEMPO = 2,2,6,6-tetramethylpiperidine-1-oxyl, DHA = 9,10-dihydroanthracene.

observed, consistent with interruption of a radical chain process through interception of the proposed chain carrier, Fp*•.

To further examine the possibility of Fp*• acting as a radical chain carrier, the allenyliron complex was also exposed to substoichiometric Fp^*_2 in the presence of light (CFL or blue LED), as previously reported by Rosenblum. Although rate acceleration was observed in these cases, the interpretation of these results was complicated by the formation of Cp*Fe(CO)(η^3-1-phenylallen-1-yl) (**1a″**),[35] which was the main product of the reaction, along with the propargyliron complex as a minor product, regardless of the light source and luminosity. Exposure of **1a** to light in the absence of Fp^*_2 afforded **1a″** cleanly at low concentration, without formation of the propargyliron species. Complex **1a″** was characterized as a solution in CD_2Cl_2 and C_6D_6 by ^{1}H NMR, ^{13}C NMR (with and without ^{1}H decoupling), and IR spectroscopy (Scheme 8,

Scheme 8. Irradiation of 1a and Key Spectral Properties of 1a″[a]



[a]Conditions: **1a** (0.03 mmol), C_6D_6 (0.5 mL), 85 W CFL, 15 min. [b]Fp^*_2 (1 mol %).

top). As shown in Scheme 7, pathway II, this observation suggests an alternate $\eta^1-\eta^3-\eta^1$ isomerization mechanism mediated by decarbonylation and recarbonylation. To examine this possibility, a solution of **1a″** in C_6D_6 (>95% purity) was subjected to bubbling of CO in an NMR tube or was stirred under a CO atmosphere in a reaction vial, either in the dark or under irradiation by a CFL. Under all conditions examined, no conversion to propargyliron complex **1a′** was observed, making pathway II unlikely to be operative in the facile isomerization of **1a** to **1a′** (Scheme 8, bottom).

Finally, we examined the possibility that isomerization could be promoted by electrophilic species through two-electron pathways (Scheme 7, pathway III). To test the possibility that $(Fp^*)^+$ could promote isomerization, $[Fp^*(thf)]^+[BF_4]^-$ (15 mol %) was added to a solution of **1a** in CD_2Cl_2 and the reaction progress was monitored by ^{1}H NMR spectroscopy at 24 °C. As shown in Figure 3, a strong rate acceleration was observed in comparison to the background, with the reaction taking place at a ca. 30 times faster rate in comparison to the uncatalyzed process, as determined by the initial rate kinetics. Given measurements of the acidity of cationic alkyne-Fp* complexes,[22] Et_3NH^+ should be acidic enough to protonate **1a** to generate equilibrium concentrations of $[Fp^*(\eta^2\text{-PhC}{\equiv}\text{CMe})]^+$, the dissociation of which would generate $(Fp^*)^+$. Accordingly, a solution of **1a** in CD_2Cl_2 was treated with $[Et_3NH]^+[BF_4]^-$ (100 mol %). A significant rate acceleration was also observed in this case, though it was less dramatic than

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what was observed when $[Fp^*(thf)]^+[BF_4]^-$ was employed (Figure 3).

Control experiments showed that Et_3N, BF_4^- sources, Ag powder, and AgI did not promote the allenyl to propargyl isomerization (see the Supporting Information). Therefore, it is likely that small amounts of cationic Fp* species in the crude product and/or prolonged exposure to $[Et_3NH]^+[BF_4]^-$ during the synthesis of the allenyliron complex **1a** are responsible for the generation of variable quantities of the propargyliron complex as a side product before we re-examined our synthetic procedure.

Under a variety of conditions, no significant allenyl to propargyl isomerization was observed for allenyliron complexes derived from 1-phenyl-1-butyne or 3-hexyne. However, the regioselectivity of the coupling of allenyliron complex **7a** and diazonium salts suggests that the propargyliron complex is present in equilibrium, though the equilibrium constant for allenyl to propargyl isomerization appears to be unfavorable.[36] In our previously reported catalytic C−H functionalization chemistry,[21,22] no sign of isomerization of the postulated allenyliron intermediate was observed, presumably due to the high electrophilicity of the oxocarbenium and iminium intermediates, which intercept the catalytic allenyliron species as soon as it forms.

CONCLUSION

In summary, we investigated the [3 + 2] cycloaddition of alkyne-derived allenyliron species with a series of dipolar reagents as a means of preparing heterocyclic compounds in a regiocontrolled manner. Using this approach, we prepared a number of heterocyclic scaffolds of general interest, including dihydrofurans, dihydropyrrolones, and pyrazoles. In the case of pyrazoles, the observed regiochemical outcome differed from our initial expectations, a result that could be rationalized by isomerization of allenyliron species to their propargyliron counterparts. We investigated this process through a series of mechanistic experiments. Further synthetic applications and investigations of ligand effects are ongoing in our laboratory and will be reported in due course.

EXPERIMENTAL SECTION

General Information. Anhydrous solvents were purchased from Acros (AcroSeal packaging), Sigma-Aldrich (Sure/Seal packaging), and Frontier Scientific (J&KSeal packaging), respectively, and were transferred into an argon-filled glovebox and used as received. All reagents were purchased from Oakwood, Acros, Alfa Aesar, or Sigma-Aldrich and used as received. Compounds were purified by flash column chromatography using SiliCycle SiliaFlash F60 silica gel, unless otherwise indicated. Thin-layer chromatography (TLC) was performed on Silicycle 250 μm (analytical) or 1000 μm (preparative) silica gel plates. Compounds were visualized by irradiation with UV light or by staining with iodine/silica gel, potassium permanganate, or phosphomolybdic acid (PMA). Yields refer to isolated compounds, unless otherwise indicated. ^{1}H NMR spectra were recorded on a Bruker 400, 500, or 600 MHz spectrometer and are referenced relative to residual $CDCl_3$ proton signals at δ 7.26 ppm, CD_2Cl_2 at δ 5.33 ppm, and C_6D_6 at δ 7.15 ppm. Data for ^{1}H and ^{19}F NMR are reported as follows: chemical shift (δ, ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, ap = apparent), integration, and coupling constant (Hz). ^{13}C NMR spectra were recorded on a Bruker 400, 500, or 600 MHz spectrometer and are referenced to $CDCl_3$ at δ 77.16 ppm, CD_2Cl_2 at δ 53.84 ppm, and C_6D_6 at δ 128.39 ppm. Unless stated otherwise, the ^{13}C NMR spectra were obtained with ^{1}H decoupling. Data for ^{13}C NMR are reported in terms of chemical shift and multiplicity where appropriate. High-resolution mass spectra were obtained on a Bruker Daltonics, Inc. APEXIII 7.0 TESLA FTMS instrument (ESI) or Waters Micromass GCT Premier instrument (EI). IR spectra were recorded on a Thermo Scientific Nicolet iS5 spectrometer (iD5 ATR, diamond) and are reported in terms of the frequency of the absorption (cm^{-1}). Elemental analyses were performed by Atlantic Microlabs Inc., Norcross, GA.

One-Pot Synthesis of Iron Complex 3. In an argon-filled glovebox, a threaded reaction tube (5 mL) equipped with a magnetic stir bar was charged with $[Fp^*(thf)]^+BF_4^-$ (203 mg, 0.50 mmol, 1.0 equiv), 1-phenyl-1-propyne (125 μL, 1.0 mmol, 2.0 equiv), ethyl trifluoropyruvate (66 μL, 0.50 mmol, 1.0 equiv), and DCE (1 mL). Subsequently, $BF_3{\cdot}OEt_2$ (123 μL, 1.0 mmol, 2.0 equiv) and distilled TMPH (235 μL, 1.5 mmol, 3.0 equiv) were added in rapid succession and the reaction tube was fitted with a septum cap and removed from the glovebox. The reaction mixture was stirred for 12 h at 80 °C, and the crude material was loaded directly onto a silica gel column packed with hexanes and purified by chromatography (hexanes/EtOAc 9/1 to 4/1) to give the product **3** as an oily yellow solid that crystallized upon storage under nitrogen. Yield: 135 mg, 51%.

Alternative Synthesis of Iron Complex 3 Starting from η^1-Allenyl Pentamethylcyclopentadienyliron Dicarbonyl Complex 1a. In an argon-filled glovebox, a threaded reaction tube (5 mL) equipped with a magnetic stir bar was charged with ethyl trifluoropyruvate (99 μL, 0.75 mmol, 1.5 equiv). A solution of freshly prepared allenyliron complex **1a** (182 mg, 0.50 mmol, 1.0 equiv) in CH_2Cl_2 (0.5 mL) was then added, and the reaction tube was fitted with a septum cap and removed from the glovebox. The reaction mixture was then stirred at room temperature (24 °C) for 2.5 h. The crude mixture was concentrated *in vacuo* and purified by flash column chromatography on neutral aluminum oxide (hexanes/CH_2Cl_2 1/0 to 10/1 to 0/1) and subsequent recrystallization to provide the desired product **3** as a bright yellow solid. A crystal suitable for X-ray diffraction was obtained by carefully laying hexanes onto a CH_2Cl_2 solution of the corresponding compound. Yield: 128 mg, 48%. ^{1}H NMR (500 MHz, CD_2Cl_2): δ 7.61 (d, J = 7.0 Hz, 2H), 7.35 (t, J = 7.0 Hz, 2H), 7.30−7.27 (m, 1H), 4.31 (q, J = 7.0 Hz, 2H), 3.25−3.15 (m, 2H), 1.70 (s, 15H), 1.34 (t, J = 7.0 Hz, 3H). ^{19}F NMR (470 Hz, CD_2Cl_2): −75.9. ^{13}C NMR (125 MHz, CD_2Cl_2): δ 217.0, 216.8, 169.2, 151.4, 134.7, 129.1, 127.6, 127.4, 124.3 (q, J = 281 Hz), 108.0, 96.5, 83.4 (q, J = 23 Hz), 62.2, 52.7, 13.8, 9.4. IR (ATR): ν(CO) 1993, 1931, 1753 cm^{-1}. HRMS (ESI): calcd for $C_{26}H_{28}F_3FeO_5$ $[M + H]^+$ 533.1233, found 533.1232. Anal. Calcd for $C_{26}H_{27}F_3FeO_5$: C, 58.66; H, 5.11. Found: C, 58.74; H, 5.14.

One-Pot Synthesis of Iron Complex 6. In a septum-capped, argon-filled reaction tube (5 mL) equipped with a magnetic stir bar and charged with $[Fp^*(thf)]^+BF_4^-$ (203 mg, 0.50 mmol, 1.0 equiv) were placed 1-phenyl-1-propyne (125 μL, 1.0 mmol, 2.0 equiv), tosyl isocyanate (82 μL, 0.54 mmol, 1.1 equiv), and DCE (1 mL). Then, in rapid succession, $BF_3{\cdot}OEt_2$ (123 μL, 1.0 mmol, 2.0 equiv) and distilled TMPH (235 μL, 1.5 mmol, 3.0 equiv) were added. After it was stirred for 12 h at 80 °C, the crude reaction mixture was purified directly by chromatography (CH_2Cl_2/EtOAc 1/0 to 49/1) to give the product **6** as an oily yellow solid that crystallized upon storage under nitrogen. A crystal suitable for X-ray diffraction was obtained by slowly cooling a solution in hot CH_2Cl_2/hexanes (1/20 v/v). Yield: 126 mg, 44%.

Alternative Synthesis of Iron Complex 6 Starting from η^1-Allenyl Pentamethylcyclopentadienyliron Dicarbonyl Complex 7a. In an argon-filled glovebox, a threaded reaction tube (5 mL) equipped with a magnetic stir bar was charged with TsNCO (37 μL, 0.22 mmol, 1.1 equiv). A solution of freshly prepared allenyliron complex **7a** (75 mg, 0.20 mmol, 1.0 equiv) in CH_2Cl_2 (0.6 mL) was then added. The reaction tube was fitted with a septum cap and removed from the glovebox. The reaction mixture was then stirred at 40 °C for 16 h. Subsequently, the reaction mixture was concentrated *in vacuo*, and the crude material was purified by flash column chromatography on neutral aluminum oxide (hexanes/CH_2Cl_2 1/0 to 10/1 to 0/1) and subsequent recrystallization from hexanes/CH_2Cl_2 (10/1 v/v) to provide the desired product **6** as a bright yellow solid.

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Yield: 86 mg, 75%. ^{1}H NMR (400 MHz, CD_2Cl_2): δ 7.57 (br s, 1H), 7.41−7.40 (m, 1H), 7.27 (t, J = 7.2 Hz, 1H), 7.21−7.19 (m, 2H), 7.04−7.02 (m, 2H), 6.98 (brs, 1H), 6.40 (brs, 1H), 5.37 (s, 1H), 2.32 (s, 3H), 2.00 (s, 3H), 1.73 (s, 15H). ^{13}C NMR (100 MHz, CD_2Cl_2): δ 216.1, 213.7, 167.4, 143.7, 139.3, 137.9, 136.8, 129.0, 128.7, 128.22, 128.16, 127.4, 125.2, 96.8, 75.8, 21.2, 14.0, 9.5. IR (ATR): ν(CO) 1993, 1940, 1730 cm^{-1}. HRMS (ESI) Calcd for $C_{30}H_{32}FeNO_5S$ [M + H]$^+$: 574.1345. Found: 574.1340. Anal. Calcd for $C_{30}H_{31}FeNO_5S$: C, 62.83; H, 5.45. Found: C, 62.57; H, 5.41.

Synthesis of Iron Complex 6′ Starting from η^1-Allenyl Pentamethylcyclopentadienyliron Dicarbonyl Complex 7a. In an argon-filled glovebox, a threaded reaction tube (5 mL) equipped with a magnetic stir bar was charged with a solution of **7a** (98 mg, 0.26 mmol, 1.0 equiv) in dry DCE (2 mL). Tosyl isocyanate (45 μL, 0.29 mmol, 1.1 equiv) was added. The reaction tube was then fitted with a septum cap and removed from the glovebox, whereupon the reaction mixture was stirred at room temperature for 1 h. The reaction mixture was loaded directly onto a column equilibrated with hexanes/CH_2Cl_2 (1/1 v/v) and quickly purified by chromatography (hexanes/CH_2Cl_2/EtOAc 9/1/0 to 1/1/0 to 0/49/1) to afford a bright yellow amorphous solid **6′** that dried to a foam under high vacuum. Yield: 64 mg, 43%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.61−7.59 (d, J = 8.3 Hz, 2H), 7.354−7.348 (m, 3H), 7.29−7.27 (m, 4H), 2.74 (q, J = 7.6 Hz, 1H), 2.42 (s, 3H), 1.62 (s, 15H), 1.21 (d, J = 7.6 Hz, 3H). ^{13}C NMR (100 Hz): 217.3, 215.8, 183.2, 144.3, 138.1, 137.3, 137.0, 136.6, 129.7, 129.2, 127.7, 127.6, 127.3, 126.3, 96.4, 21.4, 18.0, 9.3.

Ethyl 4-Bromo-5-phenyl-2-(trifluoromethyl)-2,3-dihydrofuran-2-carboxylate (3a). To a solution of recently chromatographed iron complex **3** (13 mg, 23 μmol, 1.0 equiv) in dry CH_2Cl_2 (0.5 mL) was added NBS (4.2 mg, 23 μmol, 1.0 equiv). The mixture was stirred at 24 °C under air and monitored by thin-layer chromatography until the disappearance of the starting material (1 h) and concentrated *in vacuo*, and the crude residue was purified by preparatory thin-layer chromatography (SiO_2, hexanes/EtOAc 85/15) to give **3a**. Yield: 4.6 mg, 54%. Alternatively, **3a** could be obtained using stoichiometric Br_2 in place of NBS, though side product formation was apparent and a reduced yield of the desired product was observed. Yield: 1.3 mg, 15%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.92−7.90 (m, 2H), 7.42−7.40 (m, 3H), 4.36 (q, J = 7.2 Hz, 2H), 3.54 (s, 2H), 1.35 (t, J = 7.2 Hz, 3H). ^{19}F NMR (376 MHz, $CDCl_3$) δ −78.9. ^{13}C NMR (100 MHz, $CDCl_3$): δ 166.1, 149.7, 129.7, 128.3, 128.1, 127.3, 122.7 (q, J = 282 Hz), 86.5, 83.5 (q, J = 31 Hz), 63.3, 43.4, 13.9. HRMS (ESI) Calcd for $C_{14}H_{13}O_3BrF_3$ [M + H]$^+$: 364.9995. Found: 364.9998.

Diethyl 5-Phenyl-2-(trifluoromethyl)-2,3-dihydrofuran-2,4-dicarboxylate (3b). To a solution of recently chromatographed **3** (29.2 mg, 54.9 μmol) in dry CH_2Cl_2 (0.3 mL) were added solid K_2CO_3 (22.2 mg, 161 μmol, 2.9 equiv), EtOH (1.0 mL), and ceric ammonium nitrate (CAN) (57.4 mg, 105 μmol, 1.9 equiv). The mixture was stirred in air at 24 °C until the disappearance of the starting material (2 h). The mixture was then filtered through a plug of silica, concentrated *in vacuo*, and purified by preparatory plate chromatography (SiO_2, hexanes/EtOAc 85/15) to give **3b**. Yield: 8.5 mg, 43%. ^{1}H NMR (600 MHz, $CDCl_3$): δ 7.86−7.85 (m, 2H), 7.47−7.45 (m, 1H), 7.43−7.40 (m, 2H), 4.37 (q, J = 7.2 Hz, 2H), 4.18−4.15 (m, 2H), 3.60−3.53 (m, 2H), 1.36 (t, J = 7.2 Hz, 3H), 1.24 (t, J = 7.2 Hz, 3H). ^{19}F NMR (282 MHz, $CDCl_3$): δ −78.6. ^{13}C NMR (151 MHz, $CDCl_3$): δ 165.9, 163.5, 163.3, 131.1, 129.5, 128.1, 127.8, 122.7 (q, J = 283 Hz), 101.9, 84.6 (q, J = 31 Hz), 63.3, 60.5, 37.4, 14.2, 14.0.

Ethyl 4-Iodo-5-phenyl-2-(trifluoromethyl)-2,3-dihydrofuran-2-carboxylate (3c). To a solution of recently chromatographed iron complex 3 (77.8 mg, 0.15 mmol, 1.0 equiv) in dry CH_2Cl_2 (0.5 mL) was added I_2 (0.1 M in CH_2Cl_2, 1.5 mL, 1.0 equiv) at 0 °C. The reaction mixture was stirred at 24 °C for 2 h and concentrated *in vacuo*, and the crude residue was purified by flash column chromatography (hexanes/EtOAc85/15) to give **3c**. Yield: 25 mg, 42%. Additionally, Fp*I was recovered (34 mg, 63%). ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.62−7.59 (m, 2H), 7.37−7.35 (m, 3H), 4.34 (q, J = 7.2 Hz, 2H), 3.31 (d, J = 2.4 Hz, 2H), 1.33 (t, J = 7.2 Hz, 3H). ^{19}F NMR (376 MHz, $CDCl_3$): δ −78.8. ^{13}C NMR (100 MHz, $CDCl_3$): δ 166.2, 153.5, 129.9, 128.8, 128.2, 127.9, 122.8 (q, J = 282 Hz), 84.8 (q, J = 31 Hz), 63.2, 51.8, 47.1, 14.0.

Ethyl 2-Hydroxy-5-oxo-5-phenyl-2-(trifluoromethyl)-pentanoate (3d). To a solution of recently chromatographed iron complex **3** (37 mg, 69 μmol) in CH_2Cl_2 (0.5 mL, stored in air on the benchtop) was added trifluoroacetic acid (10.7 μL, 0.14 mmol, 2.0 equiv). The reaction mixture was stirred overnight (12 h) under air. The purplish mixture was filtered through a short plug of basic aluminum oxide and washed with CH_2Cl_2 and the filtrate concentrated *in vacuo* to give the crude residue, which was purified by preparatory plate chromatography (SiO_2, hexanes/EtOAc 85/15) to afford **3d**. Yield: 21 mg, 98%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.95 (d, J = 7.4 Hz, 2H), 7.58 (t, J = 7.4 Hz, 1H), 7.47 (t, J = 7.6 Hz, 2H), 4.46−4.31 (m, 2H), 4.00 (s, 1H), 3.28−3.19 (m, 1H), 2.95−2.87 (m, 1H), 2.53−2.46 (m, 1H), 2.43−2.36 (m, 1H), 1.35 (t, J = 7.1 Hz, 3H). ^{19}F NMR (376 MHz, $CDCl_3$): δ −78.3. ^{13}C NMR (100 MHz, $CDCl_3$): δ 198.2, 169.8, 136.4, 133.4, 128.7, 128.1, 123.5 (q, J = 285 Hz), 64.1, 31.9, 26.2, 13.9 (the quaternary carbon was not observed).

4-Iodo-3-methyl-5-phenyl-1-tosyl-1,5-dihydro-2*H*-pyrrol-2-one (6a). To a solution of recrystallized iron complex **6** (33 mg, 58 μmol, 1.0 equiv) in dry CH_2Cl_2 (0.5 mL) at 0 °C was added dropwise a solution of iodine (0.1 M in $CHCl_3$, 460 μL, 0.80 equiv) over 15 min. The mixture was stirred at 0 °C for 30 min and then at room temperature for another 2 h. The solution was chromatographed directly (hexanes/EtOAc 95/5 to the desired product and 85/15 to 4/1 to Fp*I) to afford **6a**. Yield: 12 mg, 59%. Additionally, Fp*I was recovered (9.2 mg, 53%). The yield of **6a** was much lower (7.2 mg, 31%) when 1.0 equiv each of **6a** and I_2 were used instead. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.39−7.37 (m, 3H), 7.32−7.28 (m, 2H), 7.10 (d, J = 8.0 Hz, 2H), 7.05 (d, J = 8.0 Hz, 2H), 5.53 (s, 1H), 2.36 (s, 3H), 1.93 (s, 3H). ^{13}C NMR (400 MHz, $CDCl_3$): δ 165.8, 144.8, 139.6, 135.5, 134.0, 129.3, 129.2, 128.8, 128.4, 128.0, 118.9, 71.5, 21.7, 13.4.

General Procedure for the Synthesis of *N*-Arylpyrazoles. In an argon-filled glovebox, a sealed reaction tube equipped with a magnetic stir bar was charged with diazonium tetrafluoroborate (0.30 mmol, 1.5 equiv). A solution of freshly prepared allenyliron complex (0.20 mmol, 1.0 equiv) in CH_2Cl_2 (0.5 mL) was added in rapid succession. The reaction tube was sealed and moved from the glovebox. The reaction tube was placed in an oil bath, which was preheated to 60 °C. After completion of the reaction (typically 12 h), the reaction mixture was cooled to room temperature. The crude mixture was concentrated *in vacuo* and purified by flash column chromatography to provide the desired product.

*5-Methyl-1,3-diphenyl-1H-pyrazole (**10a**).* This compound was prepared following the general procedure using iron complex **7b** (59 mg, 0.19 mmol, 1.0 equiv) and phenyldiazonium tetrafluoroborate (56 mg, 0.29 mmol, 1.5 equiv). The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 10/1). The desired product **10a** was obtained as a white solid. The regioisomeric ratio (>100:1 rr) was determined by ^{1}H NMR of the crude residue. Yield: 12 mg, 27%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.87−7.85 (m, 2H), 7.54−7.47 (m, 4H), 7.42−7.37 (m, 3H), 7.33−7.30 (m, 1H), 6.53 (s, 1H), 2.39 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 151.5, 140.2, 140.0, 133.4, 129.1, 128.6, 127.8, 127.6, 125.7, 125.0, 104.4, 12.6.

*1-(4-Methoxyphenyl)-5-methyl-3-phenyl-1H-pyrazole (**10b**).* This compound was prepared following the general procedure using iron complex **7b** (53 mg, 0.17 mmol, 1.0 equiv) and 4-methoxybenzenediazonium tetrafluoroborate (56 mg, 0.25 mmol, 1.5 equiv). The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 5/1). The desired product **10b** was obtained as a pale yellow solid. The regioisomeric ratio (>100:1 rr) was determined by ^{1}H NMR of the crude residue. Yield: 16 mg, 36%. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.86−7.84 (m, 2H), 7.43−7.38 (m, 4H), 7.31−7.28 (m, 1H), 7.00−6.99 (m, 2H), 6.50 (s, 1H), 3.86 (s, 3H), 2.33 (s, 3H). ^{13}C NMR (125 MHz, $CDCl_3$): δ 159.1, 151.1,

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140.3, 133.5, 133.1, 128.5, 127.6, 126.6, 125.7, 114.3, 103.8, 55.6, 12.4.

*5-Methyl-1-(4-nitrophenyl)-3-phenyl-1H-pyrazole (**10c**).* This compound was prepared following the general procedure using iron complex **7b** (59 mg, 0.19 mmol, 1.0 equiv) and 4-nitrobenzenediazonium tetrafluoroborate (70 mg, 0.29 mmol, 1.5 equiv). The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 10/1). The desired product **10c** was obtained as a white solid. The regioisomeric ratio (>100:1 rr) was determined by ^{1}H NMR of the crude residue. Yield: 18 mg, 35%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.38−8.35 (m, 2H), 7.87−7.85 (m, 2H), 7.81−7.77 (m, 2H), 7.45−7.41 (m, 2H), 7.38−7.34 (m, 1H), 6.61 (d, J = 0.8 Hz, 1H), 2.52 (d, J = 0.4 Hz, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 153.0, 145.9, 145.0, 140.6, 132.5, 128.7, 128.4, 125.8, 124.8, 123.9, 106.6, 13.4.

*Methyl 4-(5-Methyl-3-phenyl-1H-pyrazol-1-yl)benzoate (**10d**).* This compound was prepared following the general procedure using iron complex **7b** (53 mg, 0.17 mmol, 1.0 equiv) and 4-(methoxycarbonyl)benzenediazonium tetrafluoroborate (64 mg, 0.25 mmol, 1.5 equiv). The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 10/1). The desired product **10d** was obtained as a white solid. The regioisomeric ratio (>100:1 rr) was determined by ^{1}H NMR of the crude residue. Yield: 17 mg, 35%. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.17−8.16 (m, 2H), 7.87−7.85 (m, 2H), 7.65 (d, J = 8.5 Hz, 2H), 7.43−7.40 (m, 2H), 7.35−7.32 (m, 1H), 6.57 (s, 1H), 3.96 (s, 3H), 2.46 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 166.4, 152.2, 143.6, 140.3, 133.0, 130.6, 128.7, 128.6, 128.1, 125.8, 123.9, 105.6, 52.3, 13.1.

*3-Ethyl-5-methyl-1-(4-nitrophenyl)-1H-pyrazole (**12**).* This compound was prepared following the general procedure using iron complex **11** (49 mg, 0.19 mmol, 1.0 equiv) and 4-nitrobenzenediazonium tetrafluoroborate (70 mg, 0.29 mmol, 1.5 equiv). The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 10/1). The desired product **12** was obtained as a yellow oil. The regioisomeric ratio (9.0:1 rr) was determined by GCMS. Yield: 17 mg, 40%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.34−8.30 (m, 2H), 7.70−7.67 (m, 2H), 6.11 (s, 1H), 2.68 (q, J = 7.6 Hz, 2H), 2.44 (d, J = 0.3 Hz, 3H), 1.28 (t, J = 7.6 Hz, 3H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 156.8, 145.6, 145.1, 139.8, 124.7, 123.6, 107.9, 21.5, 13.7, 13.3. HRMS (ESI) Calcd for $C_{12}H_{14}N_3O_2$ $[M + H]^+$: 232.1081. Found: 232.1084. The connectivity was assigned by nuclear Overhauser effect spectroscopy.

Synthesis of Isopropyl 5-Methyl-3-phenyl-1*H*-pyrazole-1-carboxylate (14). In a glovebox, diisopropyl azodicarboxylate (60 μL, 0.31 mmol, 1.6 equiv) and $BF_3{\cdot}OEt_2$ (38 μL, 0.31 mmol, 1.6 equiv) were dissolved in CH_2Cl_2 (0.25 mL) in a reaction tube equipped with a magnetic stir bar. A solution of allenyliron complex **7b** (59 mg, 0.19 mmol, 1.0 equiv) in CH_2Cl_2 (0.25 mL) was added in rapid succession. The reaction mixture was stirred at 60 °C for 12 h, and the crude residue was purified by flash column chromatography (hexanes/EtOAc 30/1 to 5/1). The desired product **14** was obtained as a brown oil. The regioisomeric ratio (>20:1 rr) was determined by ^{1}H NMR of the crude residue. Yield: 17 mg, 36%. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.87−7.85 (m, 2H), 7.43−7.34 (m, 3H), 6.48 (s, 1H), 5.29−5.23 (m, 1H), 2.59 (s, 3H), 1.48 (d, J = 6.4 Hz, 6H). ^{13}C NMR (125 MHz, $CDCl_3$): δ 153.9, 150.2, 145.1, 132.0, 128.9, 128.6, 126.5, 107.7, 72.6, 21.8, 14.7. HRMS (ESI) Calcd for $C_{14}H_{17}N_2O_2$ $[M + H]^+$: 245.1285. Found: 245.1285. The connectivity was verified by an independent synthesis of the pyrazole following literature reports (see the Supporting Information).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.organomet.1c00032.

Experimental procedures, kinetic data, and spectroscopic data for iron complexes and organic products (PDF)

Accession Codes

CCDC 2054882, 2059006, and 2059267 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Author

Yi-Ming Wang − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States;* orcid.org/0000-0001-6414-0908; Email: ym.wang@pitt.edu

Authors

Jin Zhu − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States*

Austin C. Durham − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States*

Yidong Wang − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States*

James C. Corcoran − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States*

Xiao-Dong Zuo − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States*

Steven J. Geib − *Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, United States;* orcid.org/0000-0002-9160-7857

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.organomet.1c00032

Author Contributions

†J.Z. and A.C.D. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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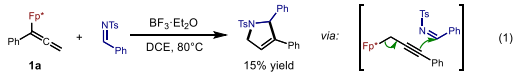
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(1)

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